

Lesson 12:

1. Designing Visuals

- **Visualization Best Practices**
 - Importance of selecting the right chart type for different types of data.
 - **Visualization of Structured Data**
 - Understanding the importance of visualizing structured datasets effectively.
 - **Choosing the Right Chart for the Data**
 - Selecting appropriate charts to draw meaningful inferences from data.
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2. Understanding Variable Types

- **Categorical Data:** Data that represents categories (e.g., Gender, Race).
 - **Binary Data:** Data with two categories (e.g., Pass/Fail, Buy/Did not buy).
 - **Ordinal Data:** Data with a ranking order (e.g., High, Medium, Low).
 - **Numerical Continuous Data:** Data that is measured on a continuous scale (e.g., Revenue).
 - **Numerical Discrete Data:** Countable data (e.g., Number of customers, vehicles).
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3. Univariate Analysis

- **Retail E-commerce Data:**
 - Example from the UCI repository of online shoppers.
 - **Types of Visuals for Univariate Analysis:**
 - **Bar Chart:** To show categorical data.
 - **Big Number:** Highlight key figures.
 - **Pie/Donut Chart:** To represent proportions in a circular format.
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4. Univariate Analysis – Extended

- **Understand Relationships:**
 - Preferred shopping days, types of visitors, and region-wise analysis.
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5. Univariate Analysis – Distribution

- **Histogram:** Represents the frequency distribution of a dataset.
 - **Frequency Distribution:** Distribution of data points across different ranges.
 - Example: **Bounce Rates:** Bounce rates for websites are between 0.000 and 0.050, with a few outliers.
 - **Conversion Rate:** Typically occurs within the first 28 minutes of the initial click (Nielsen, 2005).
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6. Univariate Analysis – Box Plot

- **Box Plot:** A visualization for distribution at quartiles.
 - **Inter-Quartile Range (IQR):** The difference between the lower and upper quartiles.
 - **Whiskers:** Extend 1.5 times the IQR. Points outside whiskers are outliers.
 - **Data Symmetry:** Shows the distribution around the median, and helps detect skewness.
 - **Business Insight:** Lower bounce rates lead to better business conversions.
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7. Multivariate Analysis

- **Data with Two or More Attributes:** Used to visualize relationships between multiple variables.
 - **Types of Multivariate Analysis:**
 - **Comparison:** Stacked bar charts, box charts.
 - **Relationships:** Scatter plots, heat maps, parallel coordinates.
 - **Trends:** Line charts, dual-axis charts.
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8. Multivariate Analysis - Comparison

- **Example:** Comparison between conversion rates on weekdays vs. weekends.
 - Weekdays: 15% conversion.
 - Weekends: 17% conversion.
- **Customer Visits:** More customers visit administration pages compared to informational pages.
 - Customers visiting administration pages are more likely to make a purchase.

9. Multivariate Analysis - Relationships

- **Scatter Plot:** Displays the relationship between two variables.
 - **Correlation vs Causation:** Not all relationships are valid; confounding or hidden variables can influence the results.
 - **Encoding:** Use color, hue, shapes, and size to represent additional variables.
- **Example:** Higher exit and bounce rates are related to low page values.
- **Joint Plot:** Shows distributions of both X and Y variables in one plot.
- **Heat Map:** Visualizes the correlation between variables.
- **Parallel Coordinates Chart:** Displays relationships between multiple variables on parallel axes.
 - Care must be taken to scale the vertical axes for accurate interpretation.
 - **Business Insight:** Page values differ for buyers and non-buyers, but there is no significant difference in exit or bounce rates.

10. Multivariate Analysis - Trends

- **Line Charts:** Show trends over time or continuous variables.
- **Dual-Axis Charts:** Example: Showing how visitors increase sales as the number of visitors increases.